SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	Y. Greeshma Reddy	**Reg. No.:**	192424146
Section / Batch:	B. Tech- AI & DS	**Date:**	29-07-2026

Instructions to Students
- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Q1. Sorting Techniques – Comparative Pass Analysis

Analyze the performance of Selection Sort and Bubble Sort on the dataset [33, 12, 27, 8, 19, 41]. Clearly interpret the problem and explain the working principle of both algorithms. Apply both methods step-by-step and show the intermediate arrays after each pass. Count the total number of comparisons and swaps in each method. Analyze their time complexities and compare their operational behavior. Interpret the results and justify which method is more suitable for this dataset.

Code of Conduct
I Y. Greeshma Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: Y. Greeshma Reddy

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				

Total Marks (100):	

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ________	Signature: ________	Signature: ________
Date: ________	Date: ________	Date: ________

Y. Greeshma Reddy

192424146

CSA0614 - DAA

CO2_AT1 - Analytical Questions Solving.

1. Sorting Techniques - Comparative Pass Analysis

a. Understanding of Problem context.

- The given dataset is [33, 12, 27, 8, 19, 41].
- The Objective is to Sort the Elements in ascending order using Selection Sort & Bubble Sort.
- The performance of both Algorithms is compared based on passes, Comparisons, Swaps & Execution behaviour.
- Intermediate arrays after every pass are analyzed to understand how each algorithm works.
- The final Comparison Identifies the most Suitable algorithm for the given dataset.

b. Application by concepts.

* Selection Sort repeatedly Selects the Smallest element from the unsorted portion & places it in its correct position.
* Bubble Sort repeatedly compares adjacent Elements & Swaps them if they are in the wrong order.
* Both Algorithms use nested Loops & perform comparisons during every pass.
* Time complexity concepts are applied to Compare Algorithm Performance.

c. Problem solving approach

* Start with the unsorted array.
* Apply Selection Sort pass by pass & record arrays.
* Count the total comparisons & swaps performed.
* Repeat the same procedure using Bubble Sort.
* Compare the final results.

d. Accuracy of Solution:

- Final sorted array: [8, 12, 19, 27, 33, 41]
- Selection Sort: 15 comparisons, 3 swaps.
- Bubble Sort: 15 comparisons, 7 swaps.
- Both Algorithms correctly sort the array.
- Selection Sort is better

e. Clarity

* Both Algorithms correctly sort the array.
* Selection Sort uses fewer swaps.
* It is more suitable for this dataset.